

Mattia Sabella

AI RESEARCHER

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 Milan

SKILLS

 Programming Languages: Python | C++ | C

 AI & Machine Learning: TensorFlow | Pytorch | Scikit-Learn | Langchain | Flower (Federated Learning)

 Data Engineering & Analytics: Pandas | NumPy | SQL | MongoDB

 Cloud & MLOps: Azure | Databricks | Git | FastAPI | Streamlit

 Software Development: Unity | Vulkan | Laravel

EXPERIENCE

POLITECNICO MILANO

PhD Student

 05/2025 - Present

 Milan, Italy

Data-Centric AI & Federated Learning

- Conduct research on Data Quality and Data Preparation impact on Machine Learning tasks
- Contributing to European Projects: BETTER & UNICA (<https://www.better-health-project.eu>, <https://www.unica-project.eu>)
- Conduct research on Federated Learning and Data Quality, contributing to the development of decentralized AI solutions

Teachings

- Teaching Experience: Sistemi Informativi (10h), Tecnologie Digitali (20h), Fondamenti di Informatica (16h)

ACCENTURE

Junior Data Scientist

 02/2024 - 05/2025

 Milan, Italy

AI Virtual Assistant for business operations

- Transformed business operations to improve client management with a chatbot for an Italian banking group.
- Developed a multi-agent system combined by:
 - a Retrieval-Augmented Generation (RAG) pipeline to extract insights from a knowledge base containing clients financial statements.
 - An SQL agent that translates natural language queries into queries for Azure SQL databases.
- Improved employees productivity and operational efficiency while increasing generative AI tools adoption within the banking sector.

Automated Data Extraction Pipeline for Loan Process Optimization

- Addressed inefficiencies in manual data extraction by automating customer data processing for loan deliberation.
- Developed a Named Entity Recognition (NER) pipeline capable of processing different document formats and extracting over 100 distinct entities using AI-powered techniques.
- Accelerated the loan approval process with automatic extraction of customer entities with 97% accuracy on monthly volume applications.

Evaluation of Large Language Models Applications

- Executed tests and evaluation metrics comparisons on LLMs to assess their NLP capabilities across multiple tasks, including translation, summarization, arithmetic reasoning, SQL translation, Named Entity Recognition, and fairness assessment.
- Provided data-driven insights to support Data Science Team and improved decision making on model selection.

EDUCATION

 M.Sc. Computer Science and Engineering

 Politecnico di Milano

 2021-2023

 102/110

 Relevant Courses: Machine Learning | Neural Networks and Deep Learning | Streaming Data Analytics | Big Data | Data Quality | Recommender Systems

 Bs.C Computer Engineering

 University of Catania

 2017-2020

 109/110

 Relevant Courses: Data Base | Web Programming | Computer Programming | Object Oriented Programming | Operative Systems | Control Systems

LANGUAGES

 Italian *Native*

 English *Proficient*

 German *Beginner*